

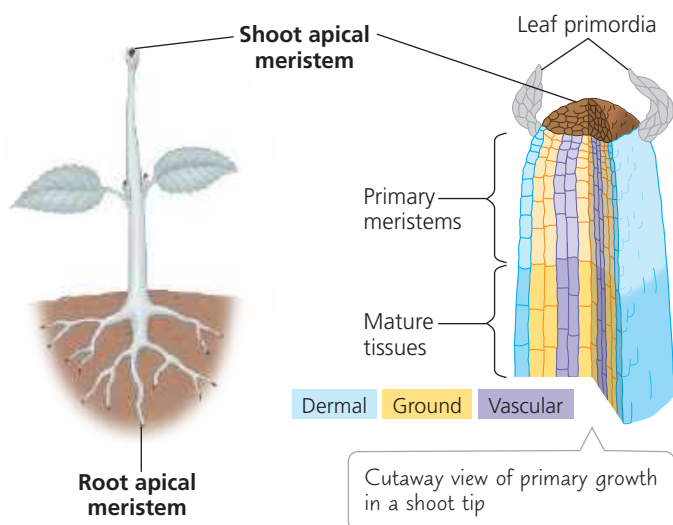
▼ Figure 35.11 VISUALIZING PRIMARY AND SECONDARY GROWTH

All vascular plants have primary growth: growth in length. Woody plants also have secondary growth: growth in thickness. As you study the diagrams, visualize how shoots and roots grow longer and thicker.

➔ **Mastering Biology Animation:**
Primary and Secondary Growth

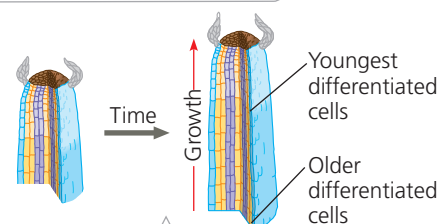
Primary Growth (growth in length)

Primary growth is made possible by apical meristems at the tips of shoots and roots.



Apical meristem cells are undifferentiated. When they divide, some daughter cells remain in the apical meristem, ensuring a continuing population of undifferentiated cells. Other daughter cells become partly differentiated as primary meristem cells. After dividing and growing in length, they become fully differentiated cells in the mature tissues.

Cell division in apical meristem
 Daughter cell in primary meristem
 Cell division in primary meristem
 Growing cells in primary meristem
 Differentiated cells (for example, vessel elements)



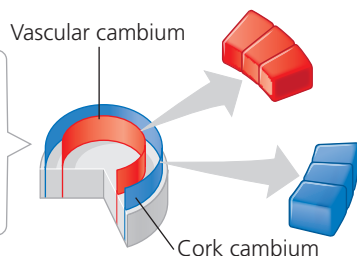
The addition of elongated, differentiated cells lengthens a stem (as shown here) or root.

1. A thimble-like root cap protects each root apical meristem. Draw and label a simple outline of a root divided into four sections: root cap (bottom), root apical meristem, primary meristems, and mature tissues.

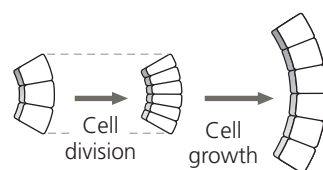
Secondary Growth (growth in thickness)

Secondary growth is made possible by two lateral meristems extending along the length of a shoot or root where primary growth has ceased.

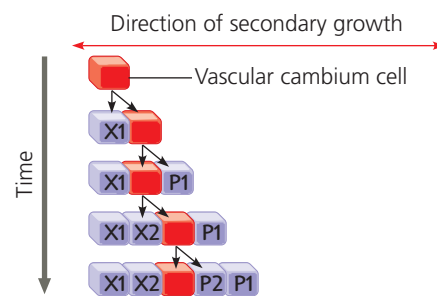
The **lateral meristems**, called the vascular cambium and cork cambium, are cylinders of dividing cells that are one cell thick.



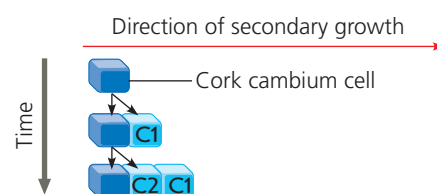
Increased circumference: When a cambium cell divides, sometimes both daughter cells remain in the cambium and grow, increasing the cambium circumference.



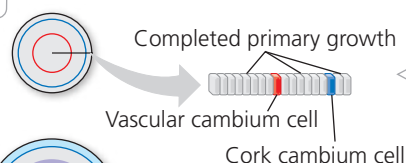
Addition of secondary xylem and phloem cells: When a vascular cambium cell divides, sometimes one daughter cell becomes a secondary xylem cell (X) to the inside of the cambium or a secondary phloem cell (P) to the outside. Although xylem and phloem cells are shown being added equally here, usually many more xylem cells are produced.



Addition of cork cells: When a cork cambium cell divides, sometimes one daughter cell becomes a cork cell (C) to the outside of the cambium.

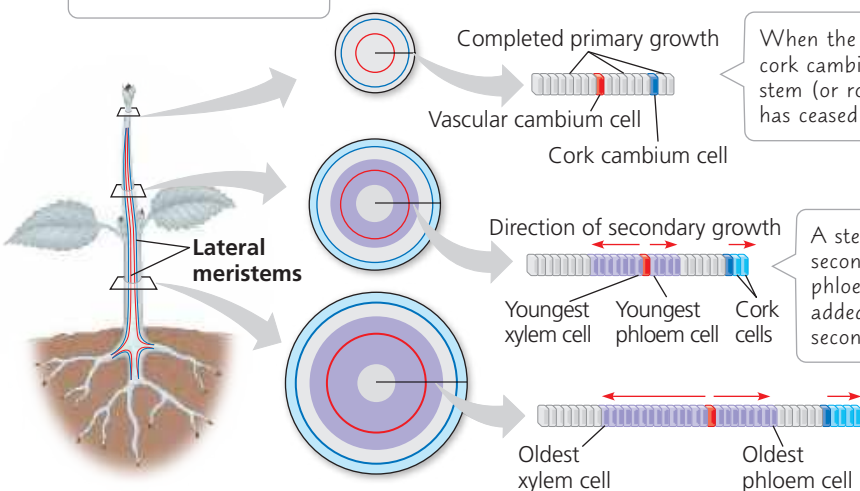


Completed primary growth: When the vascular cambium and cork cambium become active in a stem (or root), primary growth has ceased in that area.

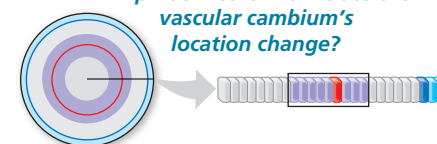


When the vascular cambium and cork cambium become active in a stem (or root), primary growth has ceased in that area.

A stem (or root) thickens as secondary xylem, secondary phloem, and cork cells are added. Most of the cells are secondary xylem (wood).



2. Draw the row of cells from the boxed area below and label the vascular cambium (V), 5 xylem cells from oldest (X1) to youngest (X5), and 3 phloem cells (P1 to P3). Show what happens after growth continues by drawing and labeling a row with twice as many xylem and phloem cells. How does the vascular cambium's location change?



➔ **Instructors:** Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.